

Protocell Integration, Replication, and Error Thresholds: The First Heritable Container Gate

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August 2026

Abstract

This paper is the integration gate. A protocell claim requires bounded copying error, retained internal chemistry, coupled energy throughput, and a boundary whose persistence covaries with its internal state. The paper combines Eigen–Orgel error-threshold logic with a conservative protocell diagnostic and explicitly reports numerical fragility where it appears. That restraint is intentional: parasitic exploitation is treated only after bounded copying and container inheritance are defined (Paper 10).

Keywords: error threshold; protocell replication; quasispecies; information fidelity; origins of life.

Preprint note

This is a computational research article. The companion script produces numerical diagnostics reported in the Computational Results section. It is not a narrative review and does not claim historical proof of a unique abiogenesis pathway.

Symbol Conventions

See Paper 1, §Symbol Conventions, for the base CDFD and Greek-symbol tables. This paper introduces the replication gain Γ_{rep} , fidelity Q , and the composite bookkeeping score $\mathcal{H}_{\text{copy}}$ (§2–3).

1 Introduction

Papers 1–8 build the prerequisite ladder: thermodynamic framing, geochemical energy and transport, oscillating polymerization, autocatalytic closure, and adaptive boundaries. A protocell requires these pieces to become coupled. Replication without a boundary may preserve

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molecular information but does not necessarily create a lineage. A boundary without inherited internal chemistry can divide physically without biological heredity.

This paper asks two linked questions:

1. When does copying preserve information rather than dissolve into error?
2. When does copying affect the persistence and division of the container that carries it?

2 Template Growth and Replication Gain

For a schematic template T and monomer pool M_i ,

$$T + M_i \rightarrow TM_i, \quad R_{\text{rep}} = k_{\text{rep}} c_T c_M. \quad (1)$$

Transport-limited monomer supply can be represented phenomenologically as

$$\mathbf{J}_M = -\frac{1}{C} \nabla c_M. \quad (2)$$

A well-mixed caricature gives

$$\frac{dc_T}{dt} = k_{\text{rep}} c_T c_M - \frac{1}{CL^2} c_T = c_T \left(k_{\text{rep}} c_M - \frac{1}{CL^2} \right). \quad (3)$$

Define the replication gain

$$\Gamma_{\text{rep}} = k_{\text{rep}} c_M CL^2. \quad (4)$$

The condition $\Gamma_{\text{rep}} > 1$ is only a linearized growth condition. It is not sufficient for life.

3 Error Threshold

If per-step fidelity is q across n additions, the simplest master-sequence survival approximation is

$$Q = q^n, \quad \epsilon = 1 - Q. \quad (5)$$

Information persists only if replication gain outruns error catastrophe. A schematic compatibility condition is

$$\Gamma_{\text{rep}} Q > 1, \quad (6)$$

with the usual warning that real prebiotic copying is not a clean independent-site process [4, 2].

The CDFD hypothesis is that fidelity, monomer activation, retention, and damage are coupled to energy throughput. A useful bookkeeping score is

$$\mathcal{H}_{\text{copy}} = \Gamma_{\text{rep}} Q R_{\text{mono}} R_{\text{ret}} R_{\text{damage}}^{-1}, \quad (7)$$

where R_{mono} is activated-monomer availability, R_{ret} is retention of templates and products, and R_{damage} collects hydrolysis, photodamage, side reactions, and strand loss. This is a checklist, not a calibrated law.

4 Quasispecies Logic Versus Protocell Logic

The quasispecies threshold and the protocell threshold are related but not identical. The first asks whether a sequence distribution preserves information. The second asks whether a bounded domain inherits a coupled chemistry that affects its growth or persistence.

The protocell condition can be stated as a covariance requirement:

$$\text{Cov}(\text{copied internal state}, \text{container persistence}) > 0. \quad (8)$$

Without this covariance, selection may act on free templates or on empty containers, but not yet on a protocell lineage.

5 Integrated Protocell Scaffold

An integrated toy protocell has state variables such as

$$U = \{c_i, \Phi_e, \Phi_H, C_i, \dots\}, \quad (9)$$

evolving through transport, reaction, and constraint laws [6]. A minimal transport skeleton is

$$\frac{\partial c_i}{\partial t} = \nabla \cdot \left(\frac{1}{C_i} \nabla c_i \right) + R_i(c, \Phi_e, \Phi_H), \quad (10)$$

with potentials satisfying analogous updates:

$$\frac{\partial \Phi_e}{\partial t} = \nabla \cdot \left(\frac{1}{C_e} \nabla \Phi_e \right) + S_e, \quad (11)$$

$$\frac{\partial \Phi_H}{\partial t} = \nabla \cdot \left(\frac{1}{C_H} \nabla \Phi_H \right) + S_H. \quad (12)$$

Flux-coupled reactions can be written as

$$\mathbf{J}_e = -\frac{1}{C_e} \nabla \Phi_e, \quad \mathbf{J}_H = -\frac{1}{C_H} \nabla \Phi_H, \quad (13)$$

$$R_i = f_i(c) g(|\mathbf{J}_e \cdot \mathbf{J}_H|), \quad (14)$$

for nonnegative g .

6 Computational Results: Numerical Audit

The diagnostic for this paper combines a reduced error-threshold sweep with a bounded two-dimensional protocell update. It should be read as a numerical stress test, not as validation. The reduced diagnostic reported:

- $\Phi = 0.2$: final master-sequence fitness 0.367,
- $\Phi = 0.5$: final master-sequence fitness 0.083,

- $\Phi = 1.0$: final master-sequence fitness 0.014,
- $\Phi = 5.0$: final master-sequence fitness 1.000,
- $\Phi = 10.0$: final master-sequence fitness 1.000.

The non-monotonic low-to-intermediate response rules out a single universal critical flux in this reduced diagnostic. The correct claim is windowed: copying benefits from energy throughput only when activation, retention, and damage remain bounded.

The final release replaces the earlier unbounded explicit autocatalytic update with saturated kinetics and bounded adaptive S, M_s updates. Under those release parameters, the two-dimensional protocell diagnostic remains finite for the full run. The result should still be read conservatively: the simulated boundary protects a low internal Ψ_s region while the exterior experiences large drive, which is a stress-test picture of compartmental buffering rather than a demonstration of real protocell reproduction.

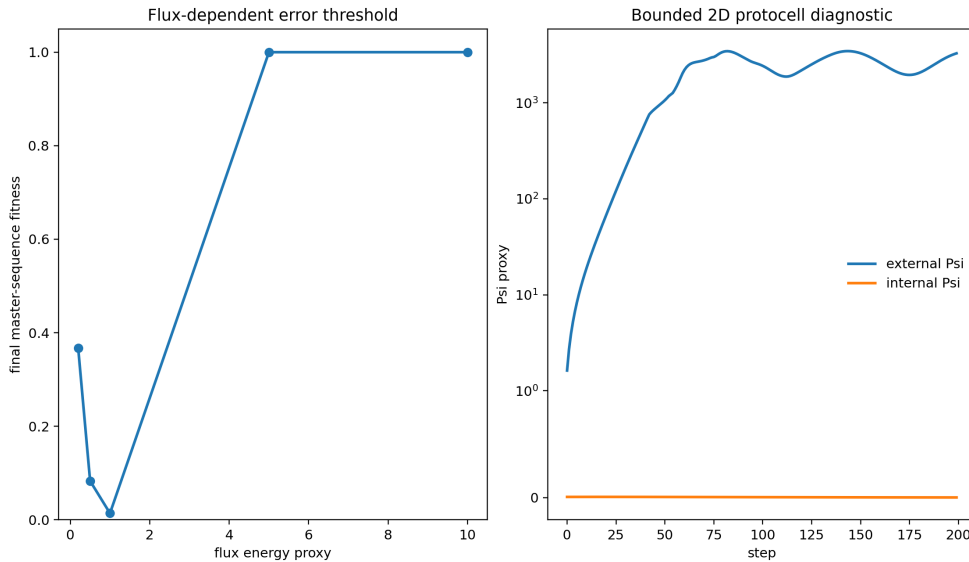


Figure 1: Merged error-threshold/protocell diagnostic. The left panel tests flux-dependent master-sequence persistence. The right panel shows a bounded two-dimensional protocell stress test with finite internal and external adaptive-ratio traces. The figure is a mechanism diagnostic, not a successful protocell simulation.

7 Hardening Roadmap

The release implementation now uses saturated kinetics such as

$$R_{\text{auto}} = \frac{k\Phi^2}{K + \Phi^2} - \ell\Phi, \quad (15)$$

and verifies boundedness in the generated summary JSON. The next hardening step is stronger: timestep refinement, grid-size refinement, and comparison against operator-split or implicit updates for coupled transport and reaction. The model should only be promoted from stress test to prediction if those refinement checks preserve the qualitative result.

8 Experimental Targets

The model protocell literature gives concrete targets: template-directed genetic-polymer synthesis in fatty-acid vesicles [3], nonenzymatic RNA synthesis inside model protocells [1], multi-step nonenzymatic primer-extension chemistry [7], and recent water-compatible thioester-mediated RNA aminoacylation/peptidyl-RNA chemistry [5]. CDFD should therefore be tested by a coupled score,

$$\mathcal{R}_{\text{proto}} = \Gamma_{\text{rep}} Q R_{\text{retention}} R_{\text{exchange}} R_{\text{division}}, \quad (16)$$

not by replication alone.

9 Conclusion

This paper is the integration gate. It establishes that bounded copying, retained internal chemistry, coupled energy flow, and heritable containers must be tested together. The current numerical status is deliberately conservative: the reduced copying toy is informative, while the two-dimensional protocell implementation is a bounded stress test that still needs refinement before it can be treated as a prediction. Parasitic exploitation is discussed only after this replication/container machinery exists (Paper 10).

10 Reproducibility Note

The companion script `supplementary_o01_09.py` writes the figure `outputs/paper_09/error_threshold_and_protocell.png` plus CSV/JSON summaries. No notebook is required.

Related literature

This manuscript is self-contained within the Part II origins-of-life sequence. Shared CDFD/CDFL symbols (Φ , C , S , M_s , Ψ_s) are used as bookkeeping only; they do not require Part I physics results. Field anchors are cited in the body and bibliography.

Competing Interests

The author declares no competing interests.

Funding

This research received no external funding.

Data and Code Availability

Companion scripts, generated figures, and numerical outputs for this manuscript are included in the public CDFD Part II release archive (<https://github.com/bampita-bico/CDFD-Part-II-Release>;

Zenodo DOI 10.5281/zenodo.20264779). See the Reproducibility Note above for the paper-local script path.

Author Contributions

Conceptualization, formal analysis, computational diagnostics, and manuscript preparation: Steve Bico Mujjabi

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